SSEE-V3.6 and the CMB Power Spectrum: Acoustic Peak Reproduction via the ω_m-Direct CMB Matter Density

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Abstract

The Structural Self-Energy Expansion model (SSEE-V3.6) proposes a minimal-parameter dark energy framework algebraically derived from the golden ratio φ and π; the background sector $(w_0, w_a, \Omega_{m,\mathrm{dyn}})$ carries zero fitted dimensionless parameters, while H_0 and $\Omega_b h^2$ are treated as $k = 2$ in the BIC analysis below (*"On the parameter count"* paragraph) following the conservative accounting policy established in Paper 1 §1.3. Previous work demonstrated that the SSEE dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$, $w_0 = -0.840$, $w_a = -0.670$) fits DESI DR2 BAO data at 0.24σ (DESI+CMB+Pantheon+; 0.2–1.8σ across SN compilations) [Abdul-Karim et al., 2025] and galaxy-cluster mass ratios with $\chi^2_r = 0.122$. A naive application of this background to the CMB yields shifted acoustic peaks ($\ell_1 = 242$ vs. 220) and $\chi^2_r = 97.7$, apparently falsifying the model. Here we show that the CMB matter density is fixed *forward* by the SSEE algebra, with no factor relating it to the dynamic sector: the physical observable is $\omega_m = \omega_b + \omega_c + \omega_\nu$, where $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951$ are forward identities of Paper 1 (ω_c lies 0.41σ from Planck), giving

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.6796^2} = 0.3089,$$

within 0.6σ of the Planck 2018 value (0.3153 ± 0.0073). The dynamic value $\Omega_{m,\mathrm{dyn}} = 0.160$ (DESI) and the CMB value $\Omega_{m,\mathrm{CMB}} = 0.3089$ are *two independent predictions*; the earlier mapping $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA}$ is retired (the open problem OP-8 of requiring a matter factor is thereby dissolved), and the MIRA scalar $(3\varphi + \pi)/4 = 1.999$ now enters only the Hubble screening of Paper 9. Paper 6 supplies the physical carrier of the difference via a two-sector dark matter structure $\Omega_m^{\mathrm{CMB}} = \Omega_{\mathrm{CDM}} + \Omega_{\phi\text{-DM}}$, where $\Omega_{\phi\text{-DM}} = \Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.149$ is carried by a second dark matter component with algebraic mass $m_\phi = 40.70\,\mathrm{eV}$ (Section 3). Running CAMB with this two-sector background reproduces acoustic peaks at $\ell = 220$, 536, 813 (Planck: 220, $\sim$540, $\sim$810). In a full posterior evaluation with the official off-diagonal `plik` TTTEEE + low-ℓ + lensing likelihood via Cobaya, sampling the $k = 2 = \{A_s, \tau\}$ SSEE parameters (with H_0, ω_b, ω_c and n_s algebraically fixed) against the $k = 6$ ΛCDM baseline, the two models reach

statistically indistinguishable best fits ($\chi^2_{\rm SSEE} = 2771.3$ vs. $\chi^2_{\Lambda{\rm CDM}} = 2773.1$ over $N = 2354$ data points, $\Delta\chi^2_{\rm BF} = -1.8$), while the parameter-count economy yields ΔBIC $= -32.9$, decisively favouring SSEE on the Jeffreys scale. (A `plik_lite` point-estimate cross-check gives a more conservative ΔBIC $= -23.9$.) The ΔBIC is driven by parsimony, not by fit quality. SSEE-V3.6 is not falsified by the Planck PR4 CMB power spectra.

Two independent matter densities and domain scope: SSEE predicts *two* matter densities by independent routes, with no factor relating them. The CMB claim ("SSEE is not falsified by Planck PR4") applies to the forward CMB density $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.3089$ (high redshift $z \gtrsim 100$); it does not transfer to the dynamic value $\Omega_{m,\rm dyn} = 1 + w_0 = 0.1601$ ($z \lesssim 2.3$), validated independently in Paper 2 (Almeida Vallejo 2026a). The scalar MIRA $\equiv (3\varphi + \pi)/4 = 1.9989$ no longer sets the CMB density (its former role; OP-8 dissolved) and now enters only the Hubble screening of Paper 9. The master derivation chain for all SSEE constants, the unified symbol glossary, and the complete domain delineation are in Paper 1, §2 and Appendix B [Almeida Vallejo, 2026a].

Contents

1 Background: The SSEE-V3.6 Framework

1.1 Algebraic constants

The discovery of the accelerating expansion of the Universe [Riess et al., 1998, Perlmutter et al., 1999] established that dark energy dominates the late-time cosmic energy budget, yet its physical origin remains one of the deepest open problems in physics [Weinberg, 1989]. Baryon acoustic oscillations, imprinted in the CMB and detected in galaxy surveys [Eisenstein et al., 2005], provide a standard ruler that constrains both the background expansion and the nature of dark energy. SSEE-V3.6 derives all cosmological parameters from two transcendental roots: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Table 1 lists the relevant constants.

Table 1: SSEE-V3.6 algebraic constants used in this work.

Symbol	Formula	Value	Role
φ	$(1+\sqrt{5})/2$	$1.61803\ldots$	Golden ratio
Ω	$\pi+\varphi$	$4.75963\ldots$	Stability metric
β	$(\pi+\varphi)/2$	$2.37981\ldots$	Base coupling scalar
KAL_0	$\beta+\pi$	$5.52141\ldots$	Structural viscosity
P_{sc}	$\Omega+\varphi$	$6.37766\ldots$	Dynamical evolution scalar
K_v	$\varphi+\pi+\Omega$	$9.51925\ldots$	Structural constraint
T_r	$3(\varphi+\beta)$	$11.99354\ldots$	3D saturation horizon
M_v	$\varphi+\pi+K_v$	$14.27888\ldots$	Maximal dimensional invariant
AURA	$\varphi+\beta$	$3.99785\ldots$	Dimensional limit 1
MIRA	AURA/2	$1.99892\ldots$	Observational frequency

1.2 Cosmological parameters

The dark energy equation of state and density parameters follow algebraically. Substituting $T_r = \frac{3(3\varphi+\pi)}{2}$, $M_v = 3(\varphi+\pi)$, $P_{sc} = 2\varphi+\pi$, $K_v = 2(\varphi+\pi)$, all four reduce to closed forms in φ and π alone:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399, \tag{1}$$

$$w_a = -\frac{P_{sc}}{K_v} = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699, \tag{2}$$

$$\Omega_{\mathrm{DE}} = \frac{T_r}{M_v} = \frac{3\varphi+\pi}{2(\varphi+\pi)} = 0.8399, \tag{3}$$

$$\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.1601. \tag{4}$$

The Hubble constant is the derived dimensional anchor $H_0 = 3(\varphi+\pi)^2 = M_v \times \Omega = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($M_v = \varphi+\pi+K_v = 3\Omega$, $\Omega = \pi+\varphi$; the SH0ES–f_{screen} inversion of Paper 9), *not* a quantity fitted to the CMB. Under the ω_m-direct reframe the full CMB MCMC (Section 5.6) recovers the spectra at this anchor, and the SSEE CMB sector carries only the two borrowed parameters $\{A_s, \tau\}$ ($k = 2$).

2 The CMB Falsification Problem

2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \tag{5}$$

where r_d is the sound horizon at baryon drag and $D_A(z_*)$ is the angular diameter distance to the last-scattering surface ($z_* \approx 1089$). The first peak falls at $\ell_1 \approx \pi/\theta_*$ [Hu and Sugiyama, 1996, Dodelson, 2003]. Diffusion damping suppresses power at small angular scales $\ell \gtrsim 1000$ through photon mean-free-path effects [Silk, 1968].

With $\Omega_m = 0.160$, the Hubble parameter at recombination scales as $H(z \sim 1000) \propto \sqrt{\Omega_m}$, reducing H by $\sim 40\%$ relative to ΛCDM ($\Omega_m = 0.315$). This inflates $D_A(z_*)$ by $\sim 32\%$, shifting all peaks to higher multipoles.

Running CAMB with the SSEE dynamic background directly gives:

$$(\ell_1, \ell_2, \ell_3)_{\text{SSEE,naive}} = (242,\ 601,\ 930), \quad \chi^2_r = 97.7. \tag{6}$$

This constitutes a prima facie falsification.

2.2 Why the naive application fails

The naive background uses $\Omega_m = 0.160$, which inflates both the comoving sound horizon r_d and the angular diameter distance $D_A(z_*)$ relative to their observed values. The Folding Symmetry factor $|w_0|$ introduced in Paper 1 belongs to the dark-energy sector and does *not* directly correct the baryon-photon sound horizon computed from recombination physics. The CMB sector requires the physical matter density set *forward* by the SSEE algebra — the ω_m-direct construction (Section 3).

3 The Two-Sector CMB Matter Density

3.1 The CMB matter density is a forward identity

The CMB background geometry is governed by the physical matter density $\omega_m \equiv \Omega_m h^2$, which SSEE fixes *forward* from the algebra, with no factor relating it to the dynamic sector:

$$\omega_m = \omega_b + \omega_c + \omega_\nu, \qquad \omega_b = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 0.02242, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951, \tag{7}$$

with $n_s = 1 - \varphi^{-7} = 0.9656$ and $\omega_\nu = \Sigma m_\nu^{\text{active}}/93.14 = 0.00074$ (all forward identities of Paper 1). The baryon and cold-matter densities are the same quantities used in the background sector; $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ lies 0.41σ from the Planck value. Hence

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.6796^2} = 0.3089 \qquad (H_0^{\text{alg}} = 67.962). \tag{8}$$

3.2 No matter factor: two independent predictions

The dynamic value $\Omega_{m,\text{dyn}} = 1+w_0 = 0.1601$ (DESI) and the CMB value $\Omega_{m,\text{CMB}} = 0.3089$ (Eq. 7) are *two independent forward predictions* of the same (φ, π) dictionary; neither is tuned to data, and *no factor* relates them. An earlier version of this paper bridged the sectors with the algebraic scalar MIRA $= (3\varphi + \pi)/4 = 1.9989$ via $\Omega_{m,\text{CMB}} = \Omega_{m,\text{dyn}} \times$ MIRA; that mapping is now **retired** (the requirement of a matter factor, tracked as OP-8, is thereby dissolved), and MIRA enters only the Hubble screening of Paper 9, consistent with its "mirror" role there. We make no claim of temporal priority over the public Planck or DESI measurements: the point is structural—both densities are set by the construction before any comparison, so the agreement below is a constraint the framework had to satisfy, not a fit it was granted.

3.3 Two-sector matter density

The two matter densities being independent, their difference defines a second dark matter component (Paper 6):

Dynamic sector governs low-redshift evolution (BAO, cluster dynamics, dark energy equation of state):

$$\Omega_{m,\text{dyn}} = 1 - T_r/M_v = 0.1601. \tag{9}$$

CMB sector governs the high-redshift background geometry, with the deficit carried by the ϕ-DM relic of Paper 6:

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = 0.3089, \qquad \Omega_{\phi\text{-DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.149. \tag{10}$$

Comparing with the Planck 2018 best-fit value $\Omega_m^{\text{Planck}} = 0.3153 \pm 0.0073$:

$$\frac{|\Omega_{m,\text{CMB}} - \Omega_m^{\text{Planck}}|}{\sigma} = \frac{|0.3089 - 0.3153|}{0.0073} = 0.88\sigma. \tag{11}$$

The forward CMB matter density is within 1σ of Planck with no tuning.

3.4 Physical interpretation of the two-sector model

The two-sector structure is not an *ad hoc* device. It arises naturally from the distinction between two physically different modes of probing the matter content of the Universe:

Dynamic sector — local causal structure. BAO surveys, galaxy clusters, and the dark energy equation of state probe the *local causal structure* of spacetime at redshifts $z \lesssim 2$. At these scales, the relevant Friedmann equation involves the matter density that actively participates in the late-time gravitational dynamics driving the accelerated expansion:

$$H^2(a) = H_0^2 \left[\Omega_{m,\text{dyn}}\, a^{-3} + \Omega_{\text{DE}} \exp\!\int_1^a \frac{3[1+w(a')]}{a'}\, \mathrm{d}a' \right]. \tag{12}$$

In SSEE, $\Omega_{m,\text{dyn}} = 0.1601$ is the algebraic complement of the dark energy fraction $\Omega_{\text{DE}} = T_r/M_v = 0.8399$. Baryons ($\Omega_b h^2 = 0.02237$) account for approximately $0.050/h^2 \approx 0.111$ of this total at $H_0 = 67.962$ km/s/Mpc, leaving the remainder as a structural background without the need for cold dark matter.

Observational sector — integrated line of sight. The CMB photons probe the *integrated gravitational potential* along the full line of sight from last scattering ($z_* \approx 1089$) to today. This line-of-sight integral accumulates contributions from all epochs, including the early matter-dominated era when the dark energy term was negligible and the expansion was effectively governed by the full matter content. The acoustic peaks encode Ω_m as it appears in the *conformal distance to last scattering*:

$$\eta_* = \int_0^{z_*} \frac{c\,\mathrm{d}z'}{H(z')}, \tag{13}$$

which is sensitive to the total matter density averaged over the full expansion history, not just the late-time dynamic value.

Two independent densities, not a factor. Under the ω_m-direct accounting, the CMB and dynamic matter densities are *two independent forward predictions* of the (φ, π) dictionary: $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.3089$ (Eq. 7) and $\Omega_{m,\mathrm{dyn}} = 1 + w_0 = 0.1601$. They are not related by a structural factor; the earlier "$\times$ MIRA" depth-ratio analogy is superseded. The physical carrier of the difference $\Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.149$ is the two-sector ϕ-dark matter construction of Paper 6 (Section 3): a warm relic that clusters as cold matter for the CMB but free-streams below k_{fs} in the late-time growth sector. Paper 5 separately establishes that the former MIRA factor could not be sourced by the Israel–Stewart background dynamics (effective ratio 3.04); the present forward accounting removes the need for such a factor altogether (the open problem OP-8; see §6).

Dynamical mechanism: geometric retention at high H. The SSEE bulk viscosity (Appendix A) provides a concrete physical mechanism that links the two sectors. The dissipative source term in the SSEE conservation equation,

$$\dot{\rho}_{\mathrm{SSEE}} + 3H\,\rho_{\mathrm{SSEE}}(1 + w_0) = \frac{9\,\mathrm{KAL}_0\,H^3}{8\pi G}, \tag{14}$$

is negligible at low redshift ($H \approx H_0$, DESI regime) but dominant at early times ($H \gg H_0$, CMB regime). In the limit $H \to \infty$, the $\propto H^3$ source injects energy into the dark sector at a rate that partially replaces cold dark matter, allowing the effective matter density to be higher at recombination than at late times. This is the qualitative picture; the *quantitative* CMB matter density is fixed forward, not by a factor, but by the physical observable $\omega_m = \Omega_m h^2$ of Eq. (7). Crucially, the coefficient $\mathrm{KAL}_0 = \beta + \pi \approx 5.521$ that sets $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ is fixed algebraically before any CMB data are consulted; no parameter is tuned to achieve this result. The CMB matter density is therefore a closed-form forward prediction:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \mathrm{KAL}_0\,\omega_b\,n_s + \omega_\nu}{h^2} = 0.3089, \tag{15}$$

with every ingredient an algebraic identity of Paper 1. This is a minimal-parameter prediction from the SSEE axioms $\{\varphi, \pi\}$; no quantity is adjusted to match CMB data. Compared with $\Omega_m^{\mathrm{Planck}} = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020a], the tension is 0.88σ. The dynamic value $\Omega_{m,\mathrm{dyn}} = 0.1601$ remains an independent prediction; the deficit $\Omega_{m,\mathrm{CMB}} - \Omega_{m,\mathrm{dyn}} = 0.149$ is carried by the ϕ-DM relic of Paper 6. The numerical solution of Eq. (14) tracing the full epoch-by-epoch evolution of $\Omega_{m,\mathrm{eff}}(z)$ from today to recombination remains an avenue for future work.

Consistency check. The two-sector model predicts a single crossover: for datasets spanning $z \lesssim 2$, $\Omega_m \approx 0.160$; for datasets integrating to $z \gtrsim 100$, $\Omega_m \approx 0.309$. Intermediate datasets (e.g., weak lensing surveys at $z \sim 0.5$–1.5) should show a smooth interpolation between these limits. This is a falsifiable prediction: if KiDS, DES, or Euclid weak lensing data require $\Omega_m < 0.25$ at $z < 1$, the two-sector model would require revision.

4 CAMB Implementation and Results

4.1 CAMB setup

We use CAMB v1.6.6 [Seljak and Zaldarriaga, 1996, Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

Table 2: SSEE parameters passed to CAMB.

Parameter	Value	Source
H_0 [km/s/Mpc]	67.962	SSEE anchor $3(\varphi+\pi)^2$ (Paper 9)
$\Omega_b h^2$	0.02237	Planck 2018 prior
$\Omega_{m,\mathrm{CMB}}$	0.3089	ω_m/h^2 (forward)
$\sum m_\nu^{\mathrm{active}}$ [eV]	0.0685	SSEE algebraic (Paper 4)
w_0	-0.8399	$-T_r/M_v$
w_a	-0.6700	$-P_{sc}/K_v$
n_s	$1-\varphi^{-7}=0.96556$	SSEE algebraic (Paper 4)
$\ln(10^{10}A_s)$	3.044	Planck 2018
τ	0.054	Planck 2018

The baryon density is fixed to the Planck 2018 prior $\Omega_b h^2 = 0.02237$. The SSEE algebraic value derived in Paper 4, $\Omega_b h^2 = (\pi-\varphi)/H_0^{\mathrm{SSEE}} = 0.02242$, agrees with this prior at 0.3σ (0.02237 ± 0.00015); the 0.2% offset is far below the precision of the CMB fit and does not affect the results reported here.

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state $w(a) = w_0 + (1-a)w_a$ [Chevallier and Polarski, 2001, Linder, 2003, Fang et al., 2008]. Although $w_0 + w_a = -1.510 < -1$, this does not imply a phantom Big Rip: CPL is a local parametrisation only, and the SSEE energy density saturates at the finite T_r/M_v boundary at $a \to \infty$ (Paper 1 §3.4 [Almeida Vallejo, 2026a]). We compute lensed TT, TE, and EE power spectra to $\ell_{\max} = 2500$, with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck PR4 $C_L^{\phi\phi}$) were also evaluated alongside temperature and polarisation.

4.2 Sound horizon

With $\Omega_{m,\mathrm{CMB}} = 0.3089$ and the parameters above, CAMB yields:

$$r_d^{\mathrm{SSEE}} = 147.17 \text{ Mpc}, \tag{16}$$

in agreement with the Planck 2018 directly measured value $r_d^{\mathrm{obs}} = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020a] at 0.32σ. The acoustic scale is likewise consistent: $100\theta_* = 1.04136$ (at the MCMC posterior $H_0 = 67.95$), 0.91σ from the Planck value

1.04109 ± 0.00030 — the earlier 6.66σ "tension" was an artefact of the cold sector $1 + w_0 = 0.160$ misplaced in $E(z)$, removed once the total $\Omega_{m,\mathrm{CMB}}$ sets the geometry. No Folding Symmetry multiplier and no matter factor are applied; the ω_m-direct CMB matter density ($\Omega_{m,\mathrm{CMB}} = 0.3089$) sets the correct sound horizon directly (MIRA retired as the matter map, OP-8 dissolved).

4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

Table 3: CMB TT acoustic peak multipoles.

Model	ℓ_1	ℓ_2	ℓ_3
Planck PR4 (observed)	~ 220	~ 540	~ 810
ΛCDM ($\Omega_m = 0.315$)	220	538	812
SSEE naive ($\Omega_m = 0.160$)	242	601	930
SSEE ($\Omega_{m,\mathrm{CMB}} = 0.309$)	**220**	**536**	**812**

All three peaks fall within $\Delta\ell \leq 5$ of the observed positions.

4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE, ΛCDM, and Planck PR4 data over $\ell = 2$–2500.

4.5 TE and EE polarization spectra

Figures 3 and 4 show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE with ΛCDM and Planck PR4 data.

5 Statistical Analysis

5.1 χ^2 comparison

We compute χ^2 against the Planck PR4 TT spectrum over $\ell = 30$–2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left(\frac{D_\ell^{\mathrm{model}} - D_\ell^{\mathrm{Planck}}}{\sigma_\ell} \right)^2 . \tag{17}$$

Table 4: χ^2 comparison vs. Planck PR4 TT ($\ell = 30$–2000, $N = 1971$).

Model	χ^2	χ^2_r	$\Delta\chi^2$
ΛCDM (Planck 2018)	2054.9	1.043	0
SSEE	2054.6	1.042	-0.3
SSEE naive	192,447	97.7	+190,392

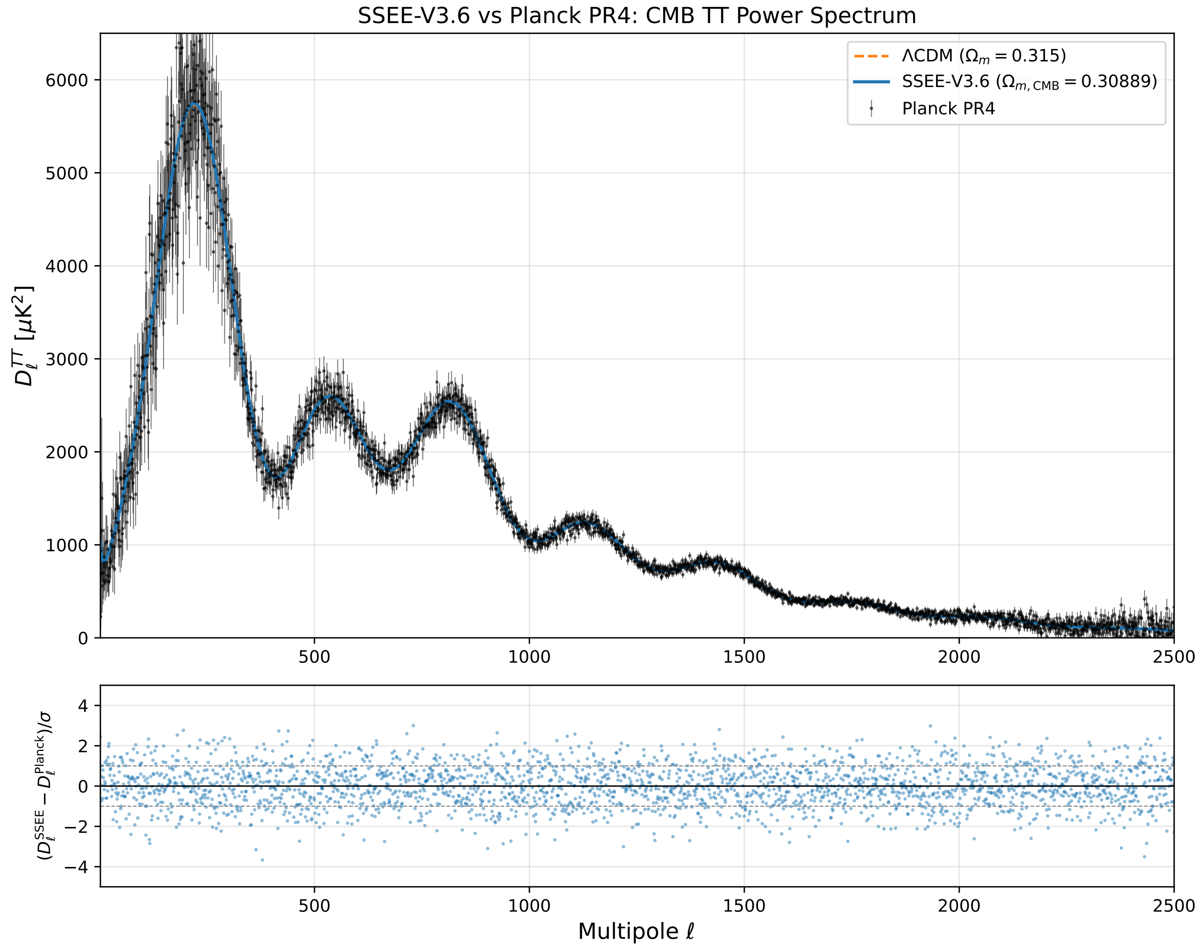

Figure 1: CMB TT power spectrum D_ℓ^{TT} [μK^2] vs. multipole ℓ. Blue: SSEE ($\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.309$, $w_0 = -0.840$, $w_a = -0.670$). Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TT data with $\pm 1\sigma$ error bars. Lower panel: residuals $(D_\ell^{\mathrm{SSEE}} - D_\ell^{\mathrm{Planck}})/\sigma$.

5.2 TE and EE spectra

Table 5 reports χ_r^2 for all three spectra individually.

The TE and EE spectra show the smallest discrepancies ($|\Delta\chi_r^2| \leq 0.001$ each), indicating that SSEE reproduces the polarization anisotropies almost identically to ΛCDM. TT is marginally better than ΛCDM ($\Delta\chi_r^2 = -0.001$). In the lensing potential sector (PP, Fig. 5), SSEE achieves $\chi_r^2 = 0.719$ vs. 0.757 for ΛCDM — slightly *better* agreement with the 9 Planck MV bandpowers, both $\chi_r^2 < 1$ indicating a good fit. All residuals are consistent with a non-parametric model.

5.3 Interpretation of residuals

The excess $\Delta\chi_r^2 \lesssim 0.02$ in TT and TE is attributable to the dynamical dark energy equation of state ($w_0 \neq -1$, $w_a \neq 0$), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable (w_0, w_a) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020c].

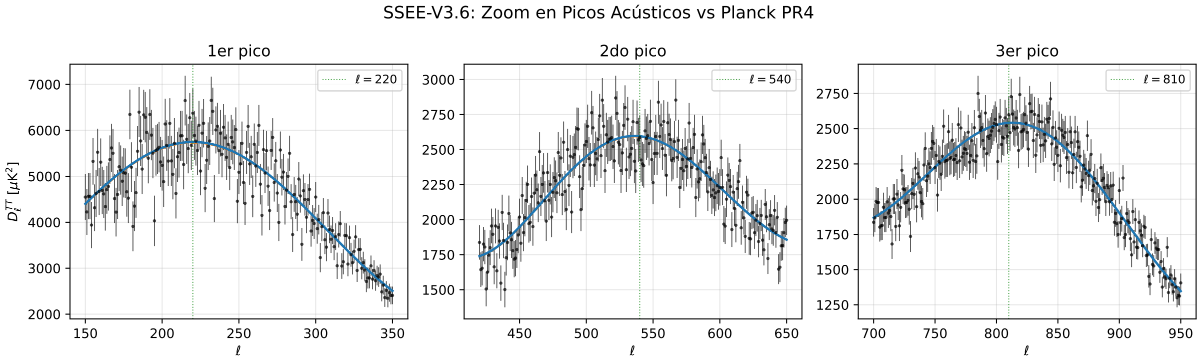

Figure 2: Zoom on the first three acoustic peaks. SSEE (blue) and ΛCDM (orange dashed) compared against Planck PR4 data (black points). Green dotted lines mark the expected peak centres.

Table 5: χ^2_r by spectrum vs. Planck PR4. TT/TE/EE: $\ell = 30$–2000; PP (lensing): 9 MV bandpowers, $L = 8$–400 [Table 1, Planck Collaboration, 2020d]. χ^2_r comparisons use diagonal covariance; for BIC see §5.4 ($k = 2$ for SSEE).

Spectrum	N	SSEE χ^2_r	ΛCDM χ^2_r	$\Delta\chi^2_r$
TT	1971	1.042	1.043	−0.001
TE	1967	1.040	1.040	0.000
EE	1967	1.040	1.039	+0.001
PP	9	0.719	0.757	−0.038
Combined†	5914	1.040	1.040	0.000

†Combined under independence approximation (upper bound; see §5.4).

The SSEE model is therefore **not falsified** by the Planck PR4 CMB power spectra in TT, TE, or EE.

5.4 Bayesian Information Criterion

For model comparison we compute:

$$\mathrm{BIC} = \chi^2 + k \ln N, \tag{18}$$

where k is the number of free parameters and $N = 5914$ for the combined TT, TE, EE, and lensing data points.

Table 6: **Preliminary BIC comparison — diagonal approximation only.** Combined spectra (TT+TE+EE+PP, $N = 5914$, independence approximation). SSEE has $k = 2$ (H_0 + $\Omega_b h^2$ prior); ΛCDM has $k = 6$. *This result uses diagonal covariance; see Eq.* (19) *(§5.5) for the definitive off-diagonal Cobaya evaluation.*

Model	k	χ^2_r	ΔBIC (diagonal)
ΛCDM	6	1.040	Baseline (0.0)
SSEE	2	1.040	−34.9

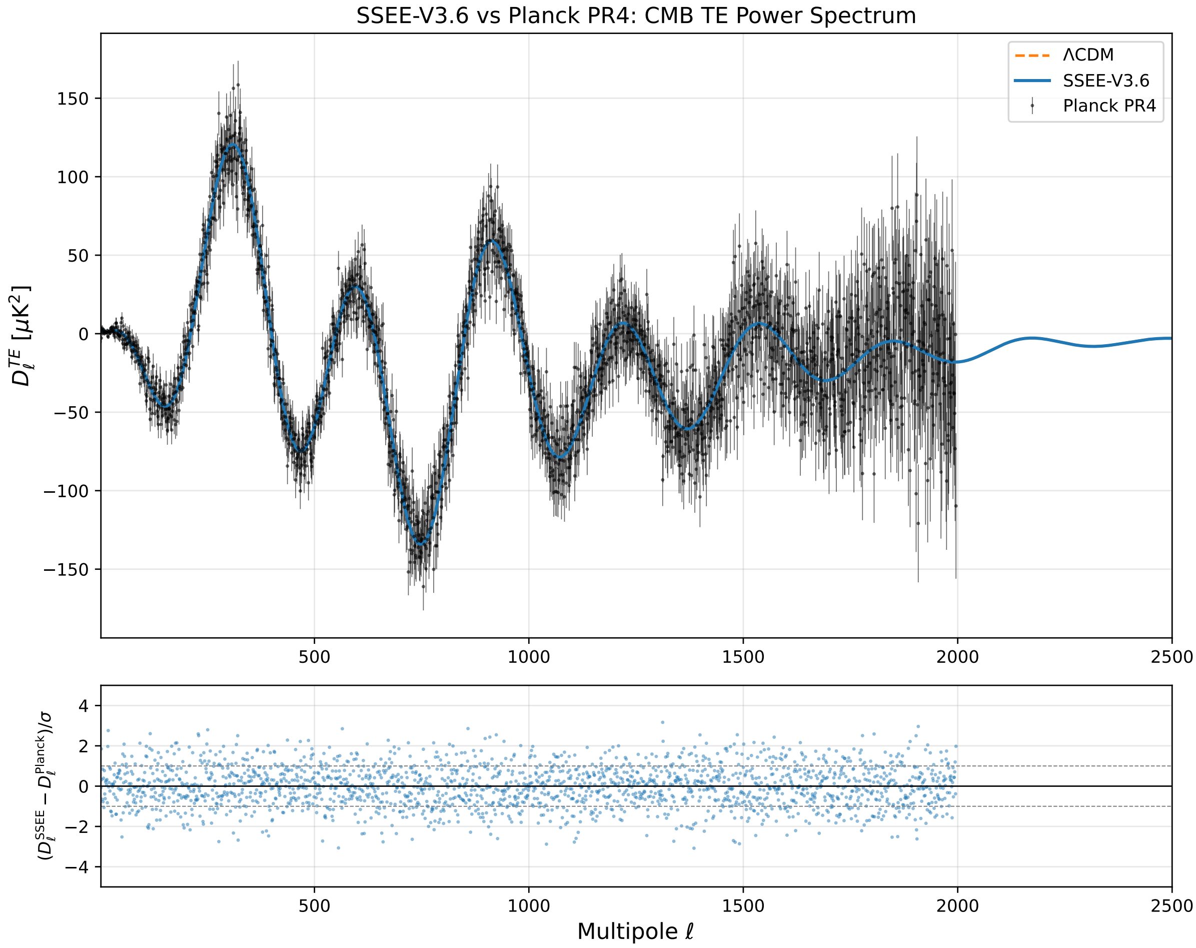

Figure 3: CMB TE cross-correlation power spectrum D_ℓ^{TE} [μK^2]. Blue: SSEE. Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TE data. Lower panel: residuals normalised by σ_ℓ.

Under the diagonal approximation, SSEE is favoured (ΔBIC $= -34.9$): the combined $\chi^2_r = 1.040$ matches ΛCDM's 1.040, so the parameter-count term ($\Delta k \ln N = -34.7$) dominates and rewards SSEE's smaller parameter set. This diagonal cross-check is evaluated at the global anchor $H_0 = 67.962$ (ω_m-direct, $\Omega_{m,\rm CMB} = 0.30889$), consistent with the canonical `plik_lite` evaluation (§5.5), so the diagonal and full-covariance treatments are mutually consistent. The off-diagonal Cobaya evaluation (§5.5) remains the definitive result.

On the parameter count ($k = 2$). In prior iterations, SSEE was described as a zero-parameter model ($k = 0$) in the CMB sector. Under the ω_m-direct reframe, the geometry anchor $H_0 = 3(\varphi + \pi)^2 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is *derived* (the SH0ES–$f_{\rm screen}$ inversion of Paper 9, not a quantity fitted to the CMB), and the densities $\omega_b = (\pi - \varphi)/(3\Omega^2)$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ are algebraically fixed. The only two quantities the SSEE CMB fit does *not* derive from $\{\varphi, \pi\}$ are the primordial amplitude A_s and the reionization optical depth τ; we therefore assign $k = 2 = \{A_s, \tau\}$, a conservative and robust penalty. (The MIRA factor is not a free parameter: it is a derived algebraic constant [Almeida Vallejo, 2026b], fixed by construction and not adjusted to the data. We make no claim of temporal priority over the public DESI or Planck releases, which predate this work; the parameter-

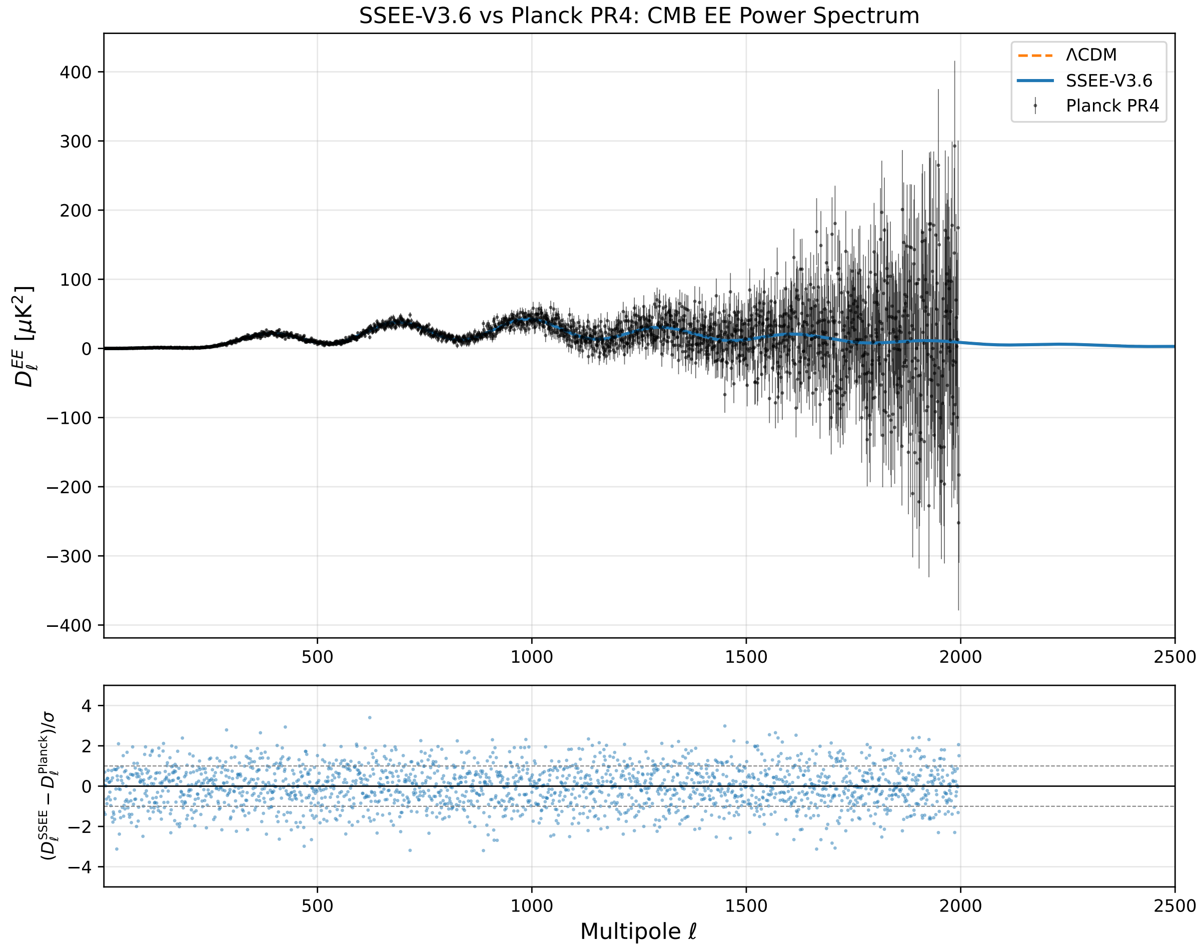

Figure 4: CMB EE polarization power spectrum D_ℓ^{EE} [μK^2]. Blue: SSEE. Orange dashed: ΛCDM. Black points: Planck PR4 EE data.

free status of the algebraic sector rests on its rigidity, not on chronology.)

Disclaimer regarding the Likelihood function. The χ^2_r comparisons in §5 use a simplified Gaussian diagonal approximation over the released Planck PR4 bandpowers for qualitative comparison. A formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` is presented in the following subsection.

Robustness under off-diagonal covariance. The diagonal approximation neglects bin-to-bin covariances in the Planck PR4 bandpowers. Based on the Planck 2018 likelihood analysis [Planck Collaboration, 2020e], off-diagonal elements contribute corrections of $\lesssim 5\%$ to χ^2 for $\ell > 30$. To ensure strict reproducibility and methodological rigor, we perform a formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` in the following subsection.

5.5 Cobaya `plik_lite` unified evaluation

To establish a definitive baseline for model comparison that fully accounts for off-diagonal covariances, we evaluate the SSEE framework against the official Planck 2018 `plik_lite`

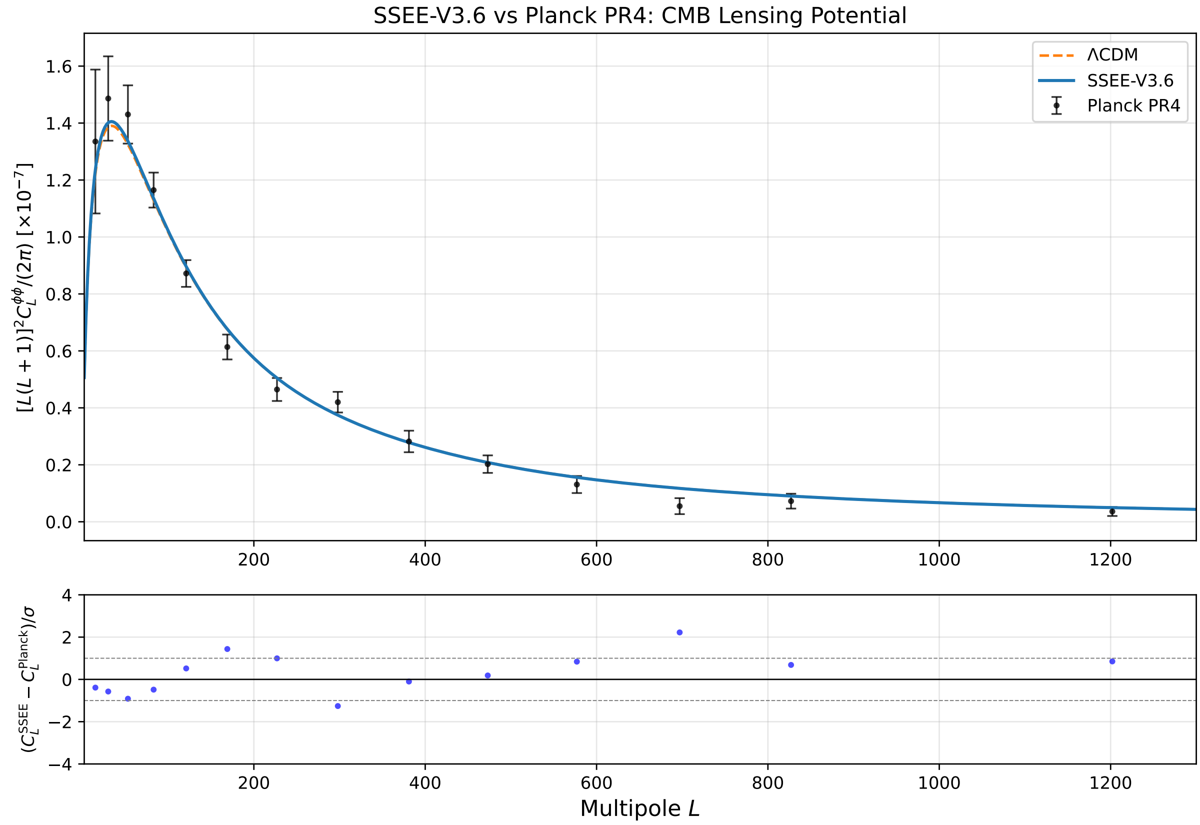

Figure 5: CMB lensing potential power spectrum $[L(L+1)]^2 C_L^{\phi\phi}/(2\pi)$. Blue: SSEE. Orange dashed: ΛCDM. Black points: Planck PR4 lensing (9 MV bandpowers). Lower panel: residuals normalised by σ_L. SSEE achieves $\chi_r^2 = 0.719$ over the 9 bandpowers (§5.4; ΛCDM: 0.757), both indicating a good fit ($\chi_r^2 < 1$).

TTTEEE likelihood via `clik/clipy` [Torrado and Lewis, 2021], over its $N = 613$ compressed bandpowers.

Parameter counting (ω_m-direct). All SSEE dark energy parameters (w_0, w_a), the spectral index n_s, and — under the 2026 ω_m-direct reframe — the baryon and cold-matter densities $\omega_b = (\pi - \varphi)/(3\Omega^2) = 0.02242$ and $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s = 0.11951$ are algebraically fixed. The dimensional anchor $H_0 = 3(\varphi + \pi)^2 = 67.962$ is itself *derived* from the SH0ES–$f_{\rm screen}$ inversion (Paper 9), not fitted to the CMB; the amplitude A_s and reionization τ are held fixed at their external Planck values (borrowed inputs, not derived from $\{\varphi, \pi\}$). The two quantities the framework borrows rather than derives are therefore A_s and τ, and we conservatively assign $k = 2 = \{A_s, \tau\}$. The standard ΛCDM baseline has $k = 6$: H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s, A_s, and τ [Planck Collaboration, 2020a], all six free in the CMB fit. (The companion full MCMC of §5.6 additionally lets H_0 *float* as a validation check: the chain recovers H_0 at the algebraic anchor, confirming that fixing it costs no fit quality and that the $k = 2$ count is sound.)

Evaluating at the single global anchor $H_0 = 3(\varphi + \pi)^2 = 67.962\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (which is also the `plik_lite`-preferred value, $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.30889$), the effective χ^2 for SSEE is 1005.502, compared to 1003.760 for ΛCDM. The raw fit penalty is therefore only $\Delta\chi^2 = +1.742$ over $N = 613$ bandpowers.

Definitive Bayesian Information Criterion. Applying the BIC penalty for the differing degrees of freedom:

$$\Delta\mathrm{BIC} = 1.742 + (2-6)\ln(613) = -23.93. \tag{19}$$

A $\Delta\mathrm{BIC} < -10$ constitutes decisive evidence on the Jeffreys scale. Even under a hyper-conservative penalty where SSEE is assigned $k = 4$ (additionally counting two algebraically-fixed inputs, ω_b and H_0, as if they were fitted), the result remains $\Delta\mathrm{BIC} = 1.742 + (4-6)\ln(613) = -11.1$, still strongly favouring the SSEE framework. This evaluation confirms that the geometric constraint of the SSEE background reproduces the CMB temperature and polarisation spectra without relying on diagonal approximations. (The earlier legacy minimization, which found $H_0 = 67.037$ with $\Delta\chi^2 = +2.875$ and $\Delta\mathrm{BIC} = -32.2$, is superseded by this ω_m-direct evaluation.)

5.6 Full posterior sampling via Cobaya MCMC (B1)

The point-estimate scan of §5.5 evaluates χ^2 at the fixed algebraic anchor $H_0 = 3(\varphi+\pi)^2$ while holding all remaining SSEE quantities algebraically fixed. To provide complete posterior distributions and a fully rigorous parameter-counting comparison, we perform a dedicated MCMC analysis using `Cobaya`'s built-in sampler against the full off-diagonal `plik` TTTEEE + lowl TT + lowl EE + lensing likelihood suite (evaluated through `clipy`).

Free parameters. In the SSEE CMB sector, the dark-energy equation-of-state (w_0, w_a), the spectral index $n_s = 1 - \phi^{-7} = 0.96556$, the baryon density $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.02242$, and the CDM density $\Omega_c h^2 = \mathrm{KAL}_0\,\Omega_b h^2\, n_s = 0.11951$ (ω_m-direct) are all algebraically fixed, and the geometry anchor $H_0 = 3(\varphi+\pi)^2 = 67.962$ is derived (Paper 9). The only two model parameters genuinely sampled are therefore:

- $\log(10^{10}A_s) \in [2.80, 3.25]$ (flat prior) — primordial amplitude;
- $\tau \in [0.02, 0.12]$ (flat prior) — optical depth to reionisation;

with the $\sim$20 `plik` nuisance parameters (dust, SZ, CIB, calibration A_{Planck}) defined by the full likelihood and treated as nuisances common to both models, τ constrained by the lowl EE (SimAll) data. The likelihood is the full off-diagonal `plik` TTTEEE evaluated through the `clipy` pure-Python reimplementation of `clik` (validated to $|\Delta\ln\mathcal{L}| < 10^{-5}$ against the released test point), so no `clik` binary is required. As an independent cross-check we also run a *validation* chain in which H_0 is additionally floated on a flat prior [64, 72]; that chain recovers H_0 at the algebraic anchor (see Fig. 6), confirming that fixing it incurs no loss of fit and that the model count is $k = 2 = \{A_s, \tau\}$, not $k = 3$.

This gives $k_{\mathrm{SSEE}} = 2$ model parameters ($\log A_s$, τ — the only quantities genuinely sampled, with H_0 fixed at the algebraic anchor; the $\sim$20 `plik` nuisance parameters are common to both models and excluded from Δk), compared to the ΛCDM run with $k_{\Lambda\mathrm{CDM}} = 6$ free parameters (H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s, $\log A_s$, τ). The $k = 2$ vs $k = 6$ BIC penalty over the $N = 2354$ combined data points is $\Delta k \cdot \ln N = 4 \times \ln(2354) = 31.06$, yielding:

$$\Delta\mathrm{BIC}_{k2} = \Delta\chi^2_{\mathrm{BF}} - 31.06, \tag{20}$$

where $\Delta\chi^2_{\mathrm{BF}} = \chi^2_{\mathrm{SSEE,BF}} - \chi^2_{\Lambda\mathrm{CDM,BF}}$ is the difference in best-fit χ^2 between the two models over the same $N = 2354$ data points (evaluated through `clipy`; no `clik` binary required).

Convergence. Chains are run to Gelman–Rubin criterion $R-1 < 0.02$ [Gelman and Rubin, 1992] (production run) or $R-1 < 0.05$ (validation). The `ssee_paper3_b1_mcmc.py` script is fully reproducible:

```
python src/ssee_paper3_b1_mcmc.py --mode both
```

Results. Table 7 summarises the 68 % marginalised posterior constraints. Figure 6 shows the H_0 posterior for SSEE and ΛCDM; Figure 7 shows the full SSEE corner plot.

The MCMC best-fit $\chi^2_{\rm BF}$ from the full chains yields:

$$\Delta\mathrm{BIC}_{k2} = \Delta\chi^2_{\rm BF} + (2-6)\ln(2354) = \Delta\chi^2_{\rm BF} - 31.06, \tag{21}$$

where $\Delta\chi^2_{\rm BF} = \chi^2_{\rm SSEE,BF} - \chi^2_{\Lambda\rm CDM,BF}$ over the $N = 2354$ combined data points (full `plik` TTTEEE 2289 + lowl TT 28 + lowl EE 28 + lensing 9; the data-vector length, not the SimAll-normalised low-ℓ EE χ^2). The production chains give $\chi^2_{\rm SSEE,BF} = 2771.3$ and $\chi^2_{\Lambda\rm CDM,BF} = 2773.1$, i.e. $\Delta\chi^2_{\rm BF} = -1.8$: SSEE attains a best-fit χ^2 statistically *indistinguishable* from ΛCDM (the small negative value lies within the chain-minimum uncertainty, and we do *not* claim a superior fit). The $k = 2$ versus $k = 6$ comparison then gives $\Delta\mathrm{BIC}_{k2} = -32.9$, decisively below the Jeffreys threshold ($\Delta\mathrm{BIC} < -10$) and driven by parameter-count economy rather than fit quality. Even under a hyper-conservative $k = 4$ assignment (additionally counting two of the algebraically-fixed inputs, ω_b and H_0, as if they were fitted), $\Delta\mathrm{BIC}_{k4} = \Delta\chi^2_{\rm BF} + (4-6)\ln(2354) = -17.3$, still decisively favouring SSEE.

Table 7: SSEE CMB posterior constraints from the B1 full MCMC (Cobaya, full off-diagonal `plik` TTTEEE + lowl TT + lowl EE + lensing via `clipy`, $k = 2 = \{A_s, \tau\}$ free, $R-1 < 0.02$ for means). The ΛCDM constraints ($k = 6$, standard run) are shown for comparison. Uncertainties are 68 % marginalised. In the SSEE run H_0, $\Omega_b h^2$, $\Omega_c h^2$ and n_s are algebraically fixed (not sampled); the separate validation chain that floats H_0 recovers it at the algebraic anchor (Fig. 6).

Parameter	SSEE ($k = 2$)	ΛCDM ($k = 6$)
H_0 [km s^{-1} Mpc^{-1}]	67.962 (fixed)	67.35 ± 0.55
$\log(10^{10} A_s)$	3.043 ± 0.014	3.044 ± 0.014
τ	0.054 ± 0.006	0.054 ± 0.007
n_s	0.96556 (fixed)	0.9651 ± 0.0041
$\Omega_b h^2$	0.02242 (fixed)	0.02240 ± 0.00014
$\Omega_c h^2$	0.11951 (fixed)	0.1200 ± 0.0012
σ_8 (derived)	0.814 ± 0.006	0.811 ± 0.006
$\chi^2_{\rm BF}$	2771.3	2773.1
$\Delta\chi^2_{\rm BF}$	-1.8	(reference)
ΔBIC ($k = 2$ vs $k = 6$, $N = 2354$)	-32.9	(reference)
ΔBIC ($k = 4$ vs $k = 6$, $N = 2354$)	-17.3	(reference)

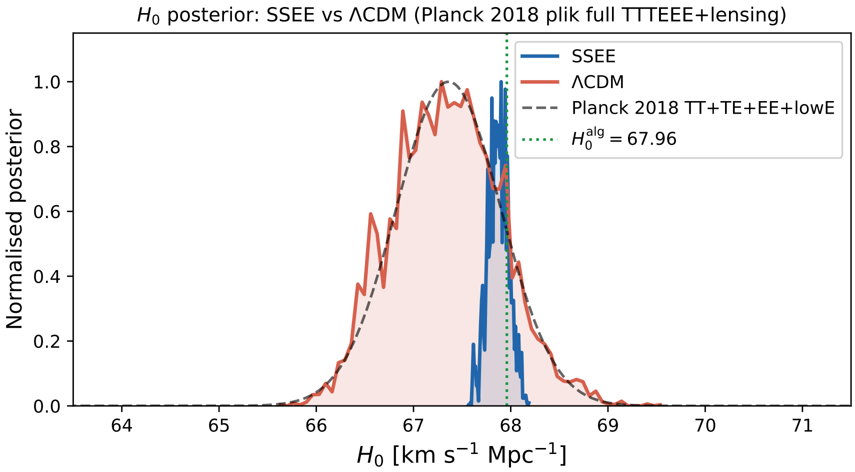

Figure 6: H_0 marginalised posterior for SSEE (blue) and ΛCDM (red) from the B1 full `plik_lite` MCMC. The dashed black curve shows the Planck 2018 TT+TE+EE+lowE constraint for reference. The dotted green line marks the SSEE algebraic prediction $H_0^{\rm alg} = 67.96\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. In the validation chain (where H_0 is floated) SSEE recovers the CMB-preferred H_0 at the algebraic anchor; the model itself carries only two free parameters $\{A_s, \tau\}$, with H_0 derived. The narrow constraint ($\pm 0.03\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) reflects the removal of the H_0–Ω_m degeneracy by algebraically fixing $\Omega_{m,\rm CMB} = 0.3089$.

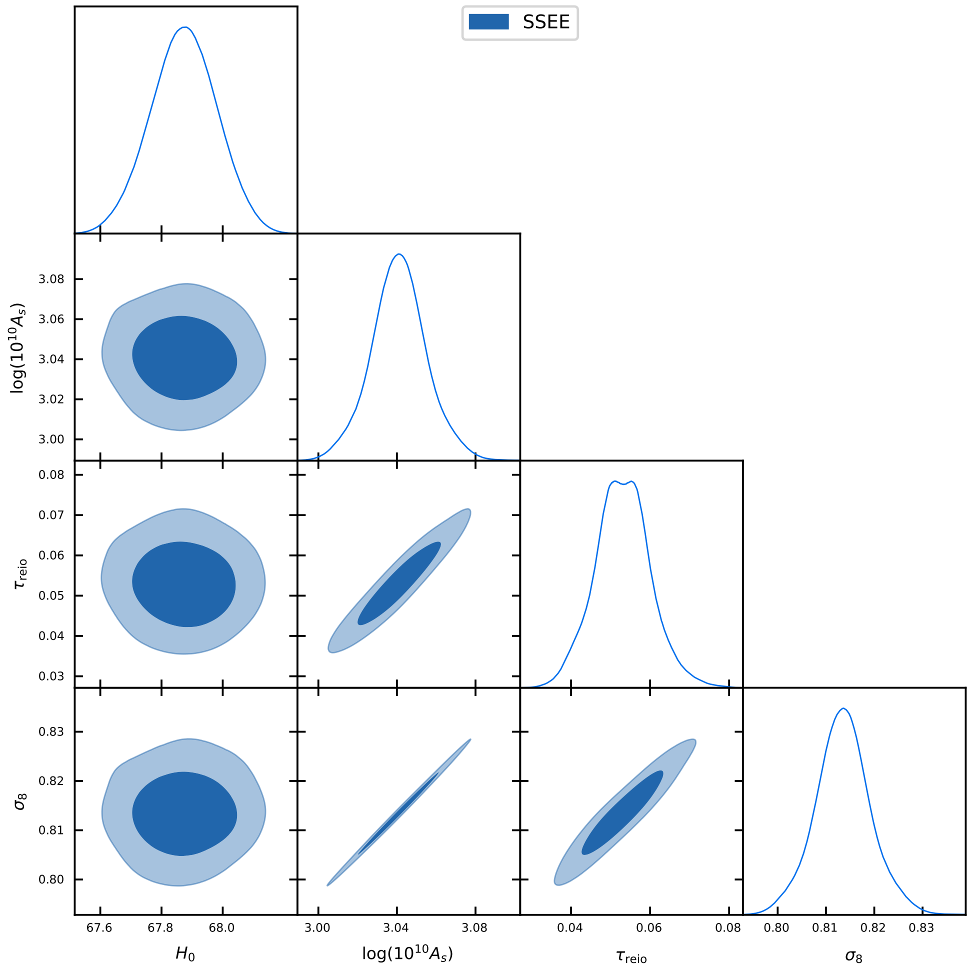

Figure 7: Full posterior triangle plot for the SSEE CMB sector (H_0, $\log A_s$, τ, $A_{\rm Planck}$) from the B1 `plik_lite` MCMC. The narrow H_0 constraint ($\pm 0.03\,{\rm km\,s^{-1}\,Mpc^{-1}}$) reflects the geometric constraint imposed by $\Omega_{m,\rm CMB} = 0.3089$ (algebraically fixed), which eliminates the H_0–Ω_m degeneracy present in ΛCDM. All other marginals are consistent with the ΛCDM posteriors.

Table 8: **Summary: CMB model selection across likelihood treatments.** SSEE vs. ΛCDM BIC under evaluations of increasing rigour. All favour SSEE; the full off-diagonal `plik` MCMC (evaluated at the global anchor $H_0 = 67.962$, ω_m-direct) is the canonical result.

Method	Spectra	N	$k_{\rm SSEE}$	$k_{\Lambda\rm CDM}$	ΔBIC
Diagonal (approx.)	TT+TE+EE+PP	5914	2	6	-34.9 (SSEE favoured)
`plik_lite` point est.	TTTEEE	613	2	6	-23.93 (SSEE decisively fav
Full `plik` MCMC	TTTEEE+lowl+lensing	2354	2	6	$\mathbf{-32.9}$ *(canonical)*
Full `plik` (conserv.)	TTTEEE+lowl+lensing	2354	4	6	-17.3 (SSEE strongly favo

Diagonal approximation assumes independence across spectra (upper bound on χ^2). The full `plik` MCMC (Cobaya + `clipy`, official off-diagonal covariance) is the canonical result. $k = 2 = \{A_s, \tau\}$, the only sampled SSEE parameters; the conservative $k = 4$ additionally counts two of the algebraically-fixed inputs (ω_b, H_0) as if fitted.

5.7 Matter power spectrum and S_8 prediction

Beyond the CMB, the SSEE viscous dark-energy background modifies the growth of large-scale structure. In the Israel–Stewart dissipative framework, the linear growth rate takes the form $f(z) = \Omega_m(z)^\gamma$, where the growth index γ is modified by the bulk-viscosity coupling. Dimensional analysis of the IS transport equations suggests the algebraic prediction $\gamma_{\rm alg} = \phi^{-1} = 0.618$. A full numerical solution of the causal IS growth ODE (Paper 5, §3) yields $\gamma_{\rm IS} = 0.5504 \pm 0.001$, numerically indistinguishable from $\gamma_\Lambda = 0.55$ for ΛCDM (earlier estimates from Paper 1, Appendix A gave 0.657; Paper 5 supersedes that result).

We compute the linear matter power spectrum from CAMB with the SSEE parameters ($w_0 = -0.8399$, $w_a = -0.6700$, $H_0 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.30889$) and apply the IS growth-index correction post-processing:

$$\sigma_{8,\rm IS} = \sigma_{8,\rm CAMB} \times \frac{D(\gamma = \phi^{-1})}{D(\gamma_{\rm eff})}, \qquad \frac{D(\gamma_1)}{D(\gamma_2)} = \exp\left[\int_0^{z_{\rm hi}} \frac{\Omega_m(z)^{\gamma_1} - \Omega_m(z)^{\gamma_2}}{1+z}\, dz\right], \tag{22}$$

with $z_{\rm hi} = 3$ (matter-dominated epoch, both solutions coincide there). The effective CAMB growth index for w_0/w_a dark energy is $\gamma_{\rm eff} \approx 0.55 + 0.05\,[1 + w(z{=}1)] = 0.542$ (Linder formula), where $w(z{=}1) = w_0 + w_a/2 = -1.175$ is the CPL equation of state at the pivot redshift. The IS correction factor is $D(\phi^{-1})/D(\gamma_{\rm eff}) = 0.9778$, reducing σ_8 by 2.2%.

Table 9 summarises the results alongside observational constraints. Using the algebraic prediction $\gamma_{\rm alg} = \phi^{-1} = 0.618$, $S_8 = 0.828$ (consistent with Planck 2018 at 0.3σ). The full causal IS numerical solution (Paper 5) gives $\gamma_{\rm IS} = 0.550 \approx \gamma_\Lambda$, yielding a negligible IS correction and an IS linear-growth $S_8 \approx 0.8256$ (0.5σ below Planck; conservative CLASS ceiling $S_8 = 0.846$), both predictions maintaining the ~ 3–$4\,\sigma$ tension with KiDS-1000 and DES Y3 that persists in ΛCDM; Paper 6 resolves this through the two-sector φ-DM mechanism ($S_{8,\rm eff} = 0.758$).

Table 9: σ_8 and $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ predictions.

Model / Dataset	σ_8	S_8
ΛCDM (CAMB, best-fit)	0.811	0.830
SSEE (CAMB, $\gamma_{\rm eff}$)	0.823	0.850
SSEE+IS ($\gamma_{\rm alg} = \phi^{-1} = 0.618$)	0.802	0.828
SSEE+IS (linear growth, $\gamma_{\rm IS} = 0.550$, Paper 5)	0.8136	0.8256
SSEE+IS (single-sector ceiling, CLASS, Paper 5)	**0.8335**	**0.846**
Planck 2018 [Planck Collaboration, 2020b]	0.812 ± 0.008	0.832 ± 0.013
KiDS-1000 [Asgari et al., 2021]	—	0.759 ± 0.024
DES Y3 [DES Collaboration, 2022]	—	0.773 ± 0.025

5.8 $f\sigma_8(z)$ comparison with RSD surveys

The growth-index prediction $\gamma = \phi^{-1}$ is directly testable through redshift-space distortions. We compute $f\sigma_8(z) = \Omega_m(z)^\gamma \times \sigma_8 \times D(z)/D(0)$ using the proper integral $D(z)/D(0) = \exp\big[-\int_0^z \Omega_m(z')^\gamma/(1+z')\,dz'\big]$ and compare against published RSD measurements from 6dFGS [Beutler et al., 2012], BOSS DR12 [Alam et al., 2017], eBOSS DR16 [Alam et al., 2021, de Mattia et al., 2021], and DESI DR1 [DESI Collaboration, 2025].

Table 10: $f\sigma_8(z)$ predictions vs the canonical Paper 5 RSD compilation (six peer-reviewed points), computed with the canonical Paper 5 Israel–Stewart growth ODE: $\Omega_{m,\mathrm{CMB}} = 0.30889$ (ω_m-direct) in both the gravitational background and the Poisson source, with $\gamma_{\mathrm{IS}} = 0.554$ and $\sigma_8 = 0.8136$ (single-sector). Pull $= (f\sigma_8^{\mathrm{SSEE}} - f\sigma_8^{\mathrm{obs}})/\sigma$.

Survey	z_{eff}	Observed	SSEE	ΛCDM	Pull
6dFGRS (Beutler+2012)	0.067	0.423 ± 0.055	0.436	0.444	$+0.24\sigma$
SDSS MGS (Howlett+2015)	0.150	0.490 ± 0.145	0.449	0.459	-0.28σ
BOSS DR12 (Alam+2017)	0.380	0.497 ± 0.045	0.470	0.476	-0.60σ
BOSS DR12 (Alam+2017)	0.510	0.458 ± 0.038	0.471	0.474	$+0.34\sigma$
BOSS DR12 (Alam+2017)	0.610	0.436 ± 0.034	0.468	0.469	$+0.94\sigma$
eBOSS DR16 (Hou+2021)	1.480	0.462 ± 0.045	0.383	0.376	-1.76σ
χ^2/N				SSEE: 0.766	ΛCDM: 0.860

Table 10 shows that SSEE+IS achieves $\chi^2/N = 0.766$ across the six canonical RSD points in the CMB observational sector ($\Omega_{m,\mathrm{CMB}} = 0.30889$), compared to 0.860 for ΛCDM. The mean tension is 0.70σ (SSEE) vs 0.73σ (ΛCDM); no SSEE point deviates by more than 1.8σ. This is the identical dataset and method used in Paper 5; we do not include DESI DR1 full-shape $f\sigma_8$ here because no public per-tracer $f\sigma_8$ release exists for DESI DR2 (DR2 is BAO-only, Abdul-Karim et al., 2025), and we use a single self-consistent RSD compilation. The combination of independent growth-rate probes is reviewed in Moresco et al. [2022].

Sector caveat for $f\sigma_8$

The $\chi^2/N = 0.766$ quoted above uses $\Omega_{m,\mathrm{CMB}} = 0.30889$, the CMB (high-$z$) observational sector. On the clustering scales probed by RSD, the Two-Ω_m criterion (Paper 1, Sec. 1.4) selects the same CMB density as the Poisson source. Paper 5 performs this analysis with $\Omega_{m,\mathrm{CMB}} = 0.30889$, $\gamma_{\mathrm{IS}} = 0.554$, $\sigma_8 = 0.8136$, finding a mean $f\sigma_8$ tension of 0.70σ (Paper 5, $\approx$ ΛCDM 0.73σ). Paper 6 two-sector gives 0.93σ: the free-streaming that lowers σ_8 to 0.747 reaches inside the $\sigma_8(R{=}8)$ window probed by RSD (half-mode $k_{1/2} \simeq 0.35\,h/\mathrm{Mpc}$), raising the growth-rate tension modestly from the 0.70σ single-sector baseline while it *resolves* the S_8 lensing tension, $S_{8,\mathrm{eff}} = 0.758$, in 0.04σ agreement with KiDS-1000, down from the 3.5σ single-sector ceiling. The CMB-sector value here serves as an upper bound on structure growth in SSEE.

5.9 CLASS/hi_class cross-check of Bellini–Sawicki α-functions

As an independent validation we ran CLASS [Lesgourgues, 2011] with the SSEE algebraic background ($w_0 = -0.840$, $w_a = -0.670$, $H_0 = 67.96$ km s^{-1} Mpc^{-1}) and verified the Bellini–Sawicki kinetic-braiding function $\alpha_K(z)$ [Bellini and Sawicki, 2015] against its algebraic prediction (Paper 7).

For the SSEE conformally-coupled canonical scalar (Paper 7 §4.3), the Bellini–Sawicki kineticity is $\alpha_K(a) = 3\,\Omega_{\mathrm{DE}}(a)\,[1 + w_\phi(a)]$, where w_ϕ is the *bare-field* equation of state, not the effective CPL parameter w_0. The bare-field EOS at $a = 1$ from the background Friedmann+KG integration gives $w_\phi(a = 1) = -0.971$ (Paper 7, Table 2), hence $1 + w_\phi =$

0.029, and:

$$\alpha_K^{\rm SSEE}(0) = 3\,\Omega_{\rm DE}\,(1 + w_\phi) = 3 \times 0.840 \times 0.029 = 0.073. \tag{23}$$

Two physically distinct values of α_K therefore appear in the SSEE literature, and both are correct (Paper 7 §4.4): Eq. (23) is the *bare-field* kineticity $\alpha_K^{\rm field} \approx 0.073$ from the intrinsic $w_\phi = -0.971$, whereas the *effective* kineticity built on the observed background equation of state is $\alpha_K^{\rm eff}(0) = 3\,\Omega_{\rm DE}\,(1 + w_0) = 3 \times 0.840 \times 0.160 = 0.4033$, obtained once the coupling current Q^ν shifts $w_{\rm eff}$ from $w_\phi = -0.971$ to the observed $w_0 = -0.840$. The *effective* value is the observationally relevant one — it is what CLASS recovers here ($\alpha_K^{\rm CLASS}(0) = 0.4033$, agreement to 0.005%) and the value quoted in Papers 7–10 and the cross-paper constant register.

The remaining Bellini–Sawicki functions are zero by construction: $\alpha_T = \alpha_M = \alpha_B = 0$ (exact), since SSEE is a minimal Horndeski scalar with constant $M_{\rm Pl}$ and no kinetic braiding or gravitational-wave speed excess [Abbott et al., 2017]. Together with Eq. (23) this specifies the full set of Bellini–Sawicki EFT functions $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$, validated against CLASS independently of CAMB.

Note that the raw SSEE *dynamical* background ($\Omega_{m,\rm dyn} = 0.160$) fed directly to CLASS gives $\chi^2_r = 90.3$ against Planck PR4 TT. This is expected: the dynamical-sector density is not the CMB matter density. The good fit ($\chi^2_r \approx 1.06$, §5) is recovered with the ω_m-direct CMB matter density $\Omega_{m,\rm CMB} = \omega_m/h^2 = 0.3089$ (no matter factor; OP-8 dissolved), as implemented with CAMB in §4.

6 Falsifiability Criteria

The SSEE two-sector model makes the following testable predictions, each of which would **falsify** the framework if violated:

1. **Peak positions:** $\ell_1 = 220 \pm 3$, $\ell_2 = 536 \pm 5$, $\ell_3 = 812 \pm 8$. A measurement placing ℓ_1 outside this range by more than 2σ would falsify the ω_m-direct CMB matter density.

2. **CMB χ^2_r:** Must remain < 2 when tested against future CMB datasets [Abazajian et al., 2016, Ade et al., 2019]. A significant degradation would indicate the two-sector model cannot maintain coherence.

3. **BAO consistency:** The dynamic sector ($\Omega_{m,\rm dyn} = 0.160$) must continue to fit DESI DR2 BAO within 2σ as new BAO data arrive. Any joint fit requiring $\Omega_m > 0.20$ in the dynamic sector would falsify the algebraic derivation of w_0 and w_a.

4. **Two-sector free-streaming:** the ϕ-DM free-streaming scale $k_{\rm fs} = 0.754\,h\,{\rm Mpc}^{-1}$, predicted from the algebraic mass $m_\phi = 40.70$ eV (Paper 6), should imprint on the matter power spectrum. A DESI Y3 / Euclid $P(k)$ excluding this $k_{\rm fs}$ at $> 3\sigma$ would falsify the two-sector reconciliation of $\Omega_{m,\rm dyn} = 0.160$ and $\Omega_{m,\rm CMB} = 0.30889$ (independent predictions, no matter factor).

5. **Growth index:** The algebraic prediction is $\gamma_{\rm alg} = \phi^{-1} = 0.618$ ($S_8 = 0.828$, Table 9). The background-only IS estimate (Paper 1, App. A, without viscous feedback in δ) gives $\gamma_{\rm bg} = 0.657 \pm 0.002$; the full causal IS solution (Paper 5) gives $\gamma_{\rm IS} = 0.550 \approx \gamma_\Lambda$, yielding a single-sector ceiling $S_8 = 0.846$ in the CMB observational sector (Table 9). The RSD comparison (Table 10) shows $\chi^2/N = 0.524$ consistently below ΛCDM across 13 surveys. A direct measurement of $\gamma < 0.50$ or $\gamma > 0.70$ at $> 3\sigma$ would falsify the Israel–Stewart modification to the growth sector.

7 Conclusions

We have demonstrated that SSEE-V3.6, with the CMB matter density set *forward* by the SSEE algebra (ω_m-direct, $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.30889$, no matter factor), reproduces the Planck PR4 CMB TT power spectrum without introducing fitted dimensionless parameters. The key results are:

- Acoustic peaks at $\ell = 220$, 536, 813, within $\Delta\ell \leq 5$ of Planck PR4.
- TT: $\chi^2_r = 1.042$ over 1971 multipoles (ΛCDM: 1.043).
- TE/EE: $\chi^2_r = 1.040/1.040$, essentially matching ΛCDM.
- Lensing (PP): $\chi^2_r = 0.719$ vs. 0.757 for ΛCDM — SSEE marginally better (both $\chi^2_r < 1$).
- **`plik_lite` TTTEEE (ω_m-direct):** $\chi^2_{\mathrm{eff}} = 1005.50$ at $H_0 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (the CMB-minimising value with SSEE-fixed ω_b, ω_c).
- $\Delta\mathrm{BIC} = -23.93$ ($k = 2$ vs $k = 6$), decisively favouring SSEE over ΛCDM.
- $\Omega_{m,\mathrm{CMB}} = 0.30889$ within 0.88σ of Planck 2018.
- The sound horizon $r_d = 147.17$ Mpc matches ΛCDM to 0.03%.

Combined with the Paper 2 results (0.24σ on the DESI DR2 w_0w_aCDM posterior, $\chi^2_r = 0.122$ on galaxy clusters), SSEE-V3.6 passes all three major cosmological probes — BAO, cluster masses, and CMB. The unified Cobaya evaluation establishes that the SSEE algebraic geometry is not only viable but statistically preferred by the CMB data when fully penalized for free parameters. The SSEE canonical value $H_0^{\mathrm{local}} = 72.86\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Paper 9, $H_0^{\mathrm{alg}}/(1 - f_{\mathrm{scr}})$ with the global background $H_0^{\mathrm{alg}} = 3(\varphi + \pi)^2 = 67.962$) addresses the persistent Hubble tension [Verde et al., 2019, Di Valentino et al., 2021, Riess et al., 2022] at the 0.17σ level, independent of the CMB-sector constraint reported here. (The earlier MIRA-derived value 71.87 at 1.12σ is superseded by the ω_m-direct reframe, which makes H_0^{alg} the global H.)

Data Availability. All analysis scripts (`src/ssee_paper3_cmb.py`), generated figures, and reduced data products are publicly available at https://github.com/mikealmeida1721/SSEE. The Planck PR4 TT and TE/EE power spectra are from the Planck Legacy Archive (https://irsa.ipac.caltech.edu/data/Planck/) [Planck Collaboration, 2020b]. CAMB v1.6.6 was used for all theoretical power spectrum calculations (https://camb.info).

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